

Hydrosaltator amictus Core Statistics

Scientific Name: Hydrosaltator amictus (The cloaked hydraulic jumper)

Common Names: Volcanic Spring Tick, Sulfur Spring Tick

Size: About 0.8 mm | Total Cell Count: ~4,000 cells

Cell allocation: ~1,400 fixed core cells; ~2,600 regenerative shell and support cells

Normal jump: about 30 mm (3 cm), or 37.5 body lengths

Lifespan: 1 to 2 active years (can survive centuries in a dormant Tun state)

Cellular Structure

- Fixed Core: brain, sensory palps, and hydraulic valves retain a fixed organization.
- Regenerative Shell: outer skin, limbs, bladder support, and pouches can slowly divide to heal.

Thermal Boundaries & Habitat

Habitat: Hydrothermal vents at depths of 2,500m to 4,960m.

- Optimal Active Range: 50C to 85C (122F to 185F).
- Below 40C triggers the dormant Tun state; above 110C is lethal.

Anatomy & Triple-Symbiosis System

The tick relies on a three-way symbiosis:

- Sugar-Silk Protist: creates extracellular matrix housing and refines sugars.
- Emberspit Microbe: harvests sulfur chemistry inside the warm matrix channels.
- Volcanic Spring Tick: distributes products and provides protected surface pockets.

The matrix is a structure created by Sugar-Silk, not a species.

Chemical Component

	Conical Pre-Wash Nozzle	Aerosolizing Blast Vent
Hyper-Saline Brine	65%	75%
Liquid Sulfur Solvent	20%	8%
Active Sulfuric/Polythionic Acid	14%	15%
Dissolved Hydrogen Sulfide Gas	1%	2%